

IAS PRELIMS 2020

STATE BY STATE



KERALA

DEMOGRAPHICAL

GEOGRAPHICAL

ENVIRONMENTAL

CULTURAL

MODULE - 13

UPSC PREVIOUS YEAR QUESTION

If you travel by road from Kohima to Kottayam, what is the minimum number of States within India through which you can travel, including the origin and the destination?

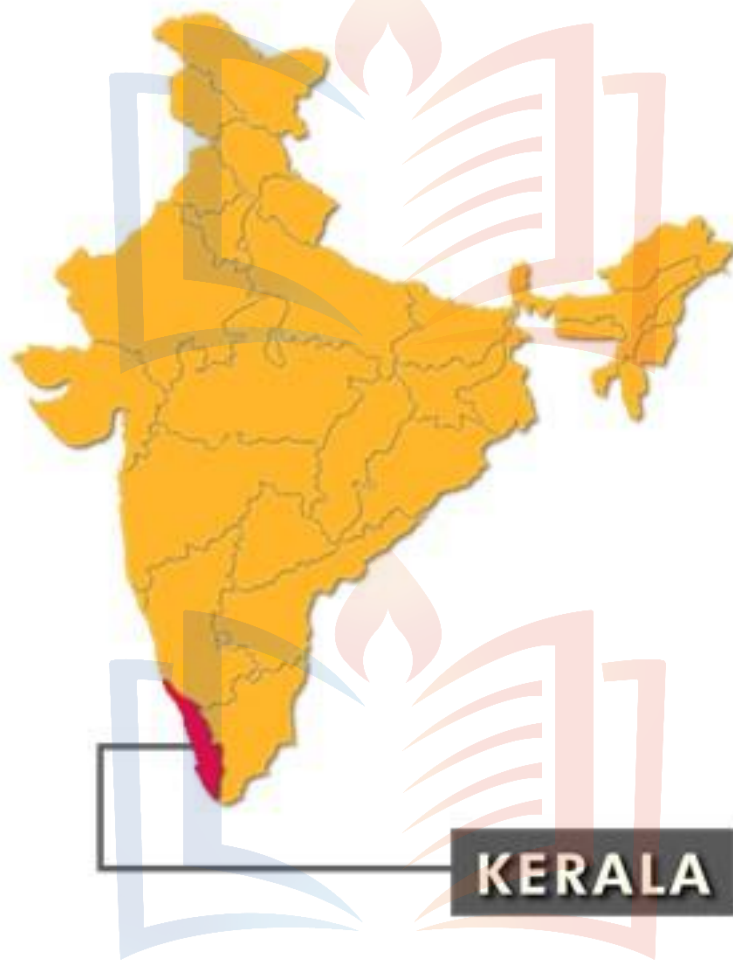
- a) 6
- b) 7
- c) 8
- d) 9



STATE BY STATE ... KERALA

1

HIGHS AND LOWS



1

Most literate state in India with a literacy rate of 93.91%.

2

At 1084 females per 1000 males, the only state which has more females than males.

3

First state in India to declare Internet access as a basic human right.

4

Ranked first in "Institutional Delivery" with almost 100% of births in medical facilities.

5

Highest life expectancy in India at 75 years.

6

With least IMR at 10 deaths per 1000 live births, it is the most baby-friendly state.

7

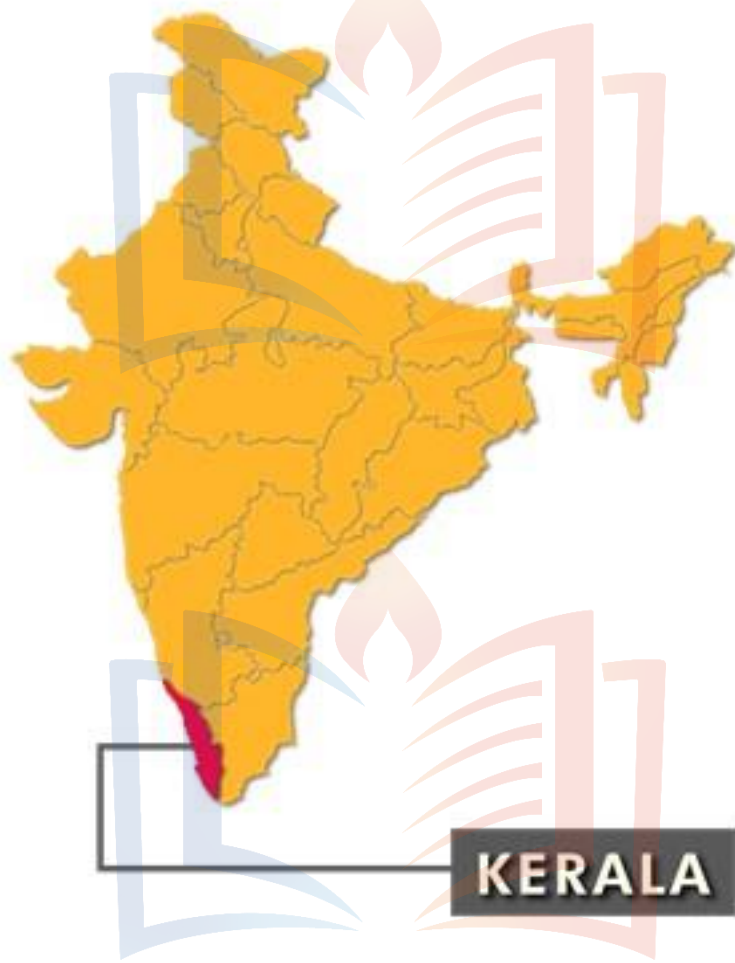
Produces more than 90% of the total rubber in the country.

8

India's first police force to induct humanoid robots in SI rank.

1

HIGHS AND LOWS



9

India's first digital state. With the state making remarkable strides in the implementation of various e-governance initiatives, it was honoured with the tag of "Complete Digital State" in 2015. (Also, remember Cyberdome Project).

10

Cochin international airport - World's first fully solar-powered airport. Awarded 'Champion of the Earth' by the UN for its contribution towards the environment.

11

Kerala's efforts to promote exclusive breastfeeding of infants for six months and supplementary breastfeeding beyond that age, made it the world's first "baby-friendly" state under the World Health Organization-UNICEF Baby-Friendly Hospital Initiative in 2002.

STATE BY STATE ... KERALA





KARNATAKA

ARABIAN SEA

KERALA

LAKSHADWEEP

- Wedged between Arabian Sea and Western Ghats.
- Geographically, Kerala can be divided into three climatically distinct regions:
 - ✓ the eastern highlands: rugged and cool mountainous terrain,
 - ✓ the central mid-lands: rolling hills, and
 - ✓ the western lowlands: coastal plains.
- Eastern region of Kerala consists of high mountains, gorges and deep-cut valleys to the immediate west of the Western Ghats' rain shadow.
- Kerala's western coastal belt is relatively flat compared to the eastern region, and is criss-crossed by a network of interconnected brackish canals, lakes, estuaries, and rivers known as the Kerala Backwaters.

3

RELIEF FEATURES

STATE BY STATE ... KERALA



Area of the Vembanadu- Kol wetlands is divided into two zones -

- ✓ the freshwater dominant zone and
- ✓ the salt water dominant zone.

STATE BY STATE ... KERALA

3

RELIEF FEATURES



- Western Ghats form a wall of mountains interrupted only near Palakkad. Hence also known Palghat, where the Palakkad Gap breaks.
- Also known as the Sahyadri mountains in Maharashtra and Karnataka, Nilagiri malai in Tamilnadu, and Sahya Parvatam in Kerala.
- Western Ghats passes through 6 states.
- UNESCO's World Heritage Site and also one of the eight "hottest hot-spots" of biological diversity in the world.



- **Periyar river:** Longest river and named as “Lifeline of Kerala”. Supreme Court in its judgement on the Mullaperiyar issue in 2014, made it clear that Tamil Nadu is not a riparian state of the river Periyar but the river itself is an inter-State river.
- **Bharathapuzha River:** Headwaters of the main tributary of Bharathapuzha originates in the Anaimalai Hills in the Western Ghats, and flows westward through Palakkad Gap.
- **Pampa/Pamba river:** Sabarimala temple dedicated to Lord Ayyappa is located on its bank. Also known as 'Dakshina Bhageerathi'.



- **Bhavani river:** Originates from Nilgiri hills of the Western Ghats. It enters the Silent Valley National Park in Kerala and flows back towards Tamil Nadu to join Cauvery river.
- **Kabini, Bhavani and Pambar rivers** rise in Kerala and flow eastwards. Kabini flows into Karnataka and the other two into Tamil Nadu.

- As Kerala's rivers are small and lacking in delta, they are more prone to environmental effects.
- Rivers face problems such as sand mining and pollution.
- State's largest lake Vembanad, dominates the backwaters. Lies between Alappuzha and Kochi.
- State experiences several natural hazards like landslides, floods and droughts. Also affected by the 2004 Indian Ocean tsunami, and in 2018 received the worst flooding in nearly a century.



- **Malampuzha:** Largest reservoir in Kerala.
- **Parambikulam:** Operated and maintained by Tamil Nadu but the ownership rests with Kerala. Parambikulam river is a tributary of Chalakudy river.
- **Idukki:** Constructed across the Periyar River in a narrow gorge between two granite hills locally known as Kuravan and Kurathi.
- **Mullaperiyar:** Located in Kerala on the river Periyar, but is operated and maintained by Tamilnadu. Periyar National Park is located around the dam's reservoir.

ARABIAN SEA

LAKSHADWEEP

	1. Forest Soil
	2. Alluvial Soil
	3. Red and Yellow Soil
	4. Black Soil
	5. Laterite Soil
	6. Arid Soil

Forest Soil

Black Soil

Laterite Soil

Coastal Alluvial Soil

Kari Soil

Alluvial Soil

RED & YELLOW



BAY OF BENGAL

- Peaty soils are locally called Kari in Kottayam and Alleppey districts of Kerala. Generally submerged during the rainy season and utilised for the cultivation of rice.
- Formation: These are marshy soils and are a result of water logging and anaerobic conditions (which leads to partial decomposition of organic matter).

STATE BY STATE ... KERALA

6

RAINFALL



Climate:

- Tropical monsoon with seasonally excessive rainfall and hot summer except over Thiruvananthapuram district, where the climate is tropical savanna with seasonally dry and hot summer weather.

Rainfall:

- Southwest monsoon (June-October) is the principal rainy season with 70% of its annual rainfall.
- Monsoon rainfall as a % of annual rainfall decreases from north to south.
- Northeast monsoon rainfall as a % of annual rainfall increases from north to south.
- Rainfall amount in the State decreases towards the south with decrease of height of Western Ghats.
- EDAVAPATHI: Thunderstorm rains in the pre-monsoon months of April and May and that of monsoon months.
- THULAVARSHAM: Rainfall during northeast monsoon season.
- Southwest monsoon sets-over the southern parts of the State by about 1st June and extends over the entire State by 5th June.

ARABIAN SEA



CROPS

KARNATAKA

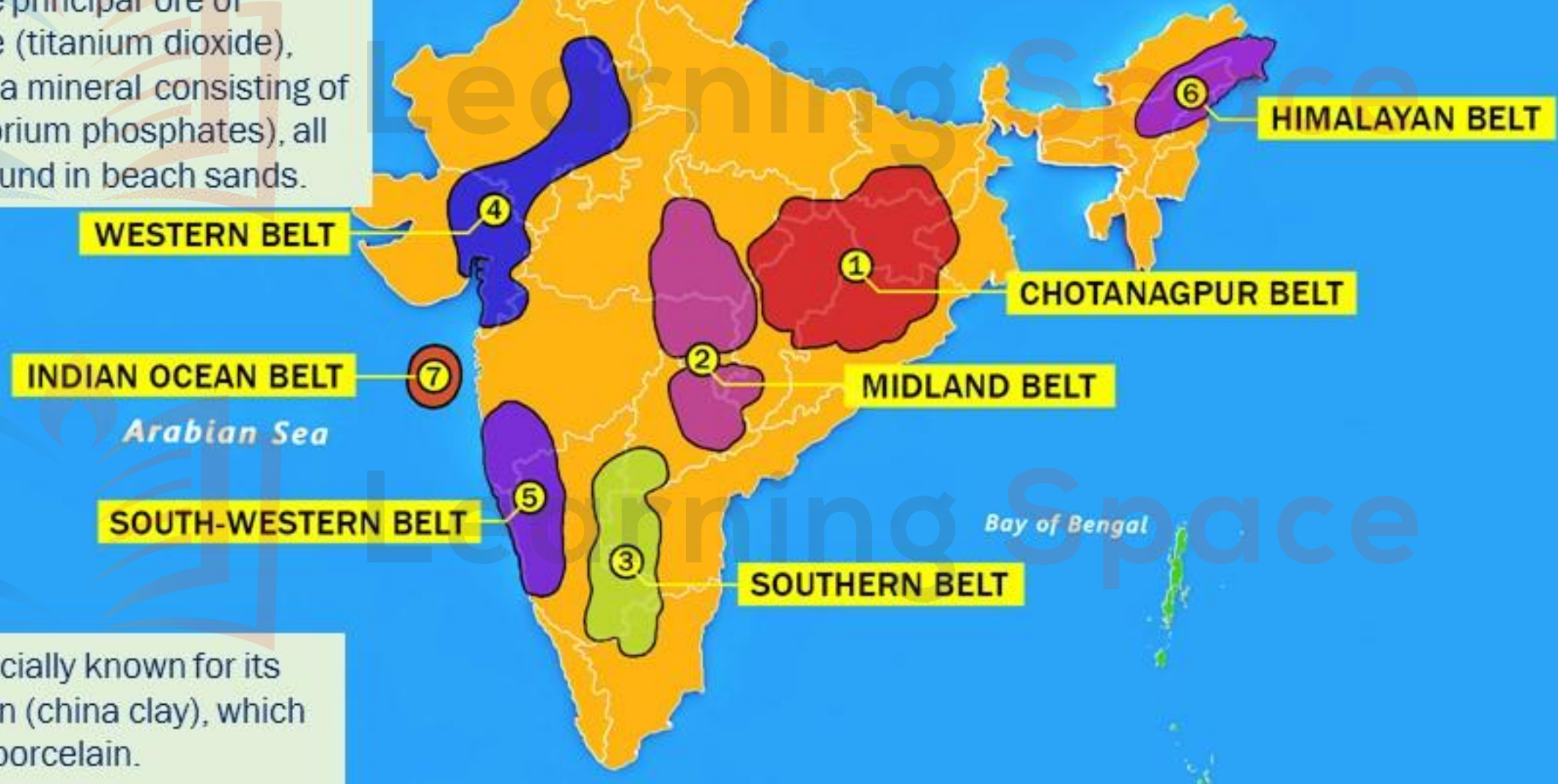
Kuttanad

- Major crops are paddy, coconut, pepper, cashew and plantation crops like rubber.
- Cash crops like coconuts, rubber, tea and coffee, cashews and sugarcane give the agriculture of Kerala a distinct flavour.
- It is coconuts which bring the people their principal source of earning in agriculture.
- Produces more than 90% of total rubber in the country. Rubber board is located at Kottayam.

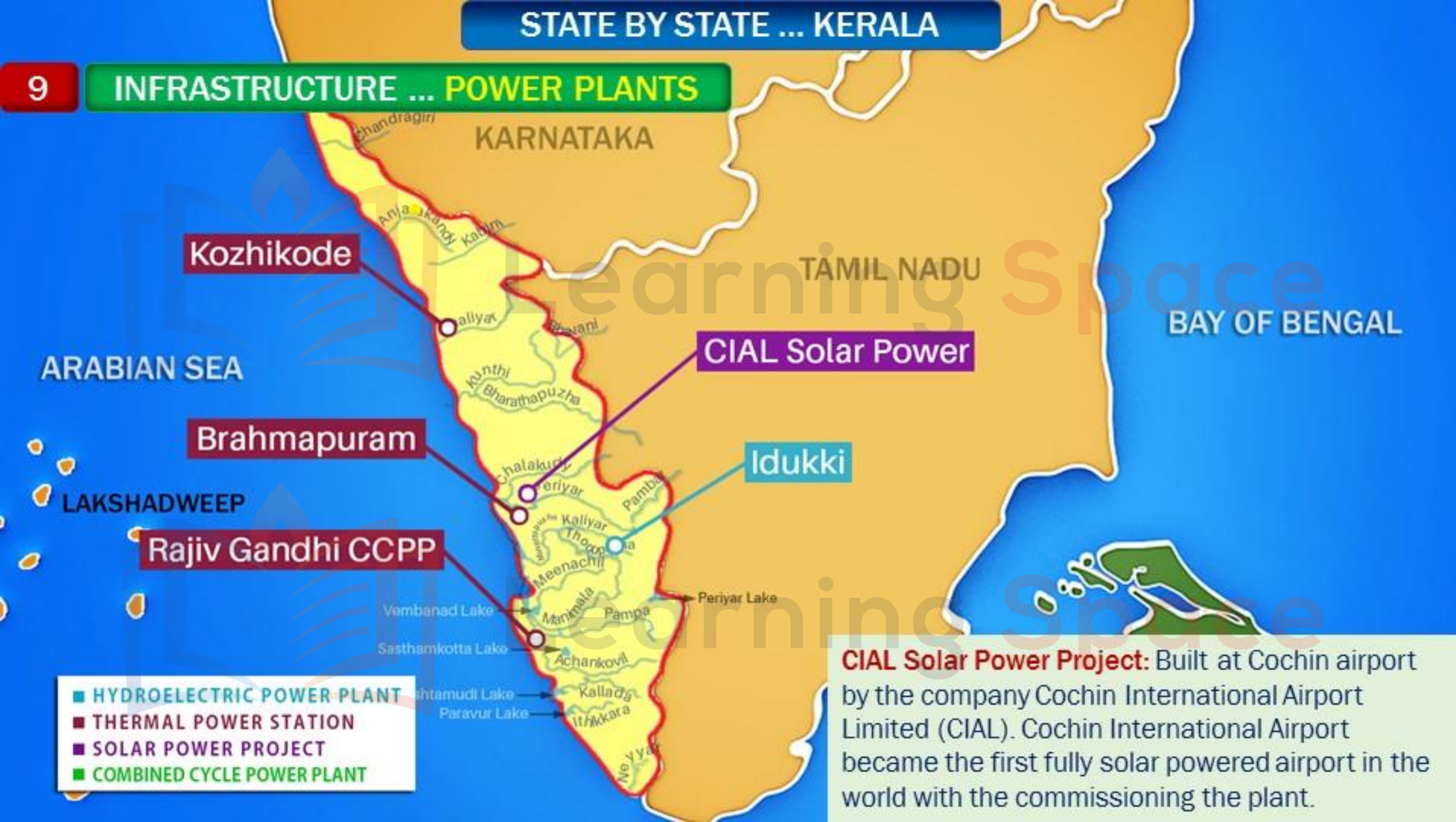
- Kuttanad Below Sea-level Farming System (KBSFS) is unique as it is the only system in India that practices rice cultivation below sea level.
- Kuttanad Wetland Agriculture System is part of Globally Important Agricultural Heritage Systems (GIAHS) designated by FAO.

8 MINERAL BELTS IN INDIA

- Lacks major reserves of fossil fuels.
- However, there are moderate deposits of ilmenite (the principal ore of titanium), rutile (titanium dioxide), and monazite (a mineral consisting of cerium and thorium phosphates), all of which are found in beach sands.



The state is especially known for its high-quality kaolin (china clay), which is used to make porcelain.



ARABIAN SEA

BAY OF E

TAMIL NADU

LAKSHADWEEP

SEA PORTS

Cochin

Kollam

Vizhinjam

Anjarakandy
Kallim

Chaliya

Bharani

Kunthi
BharathapuzhaChalaky
periyar

Pambol

Kaliyar

Thoppuzha

Meenachil

Mankala

Pampa

Vembanad Lake

Sasthamkotta Lake

Ashtamudi Lake

Paravur Lake

Achanthol

Adakkara

Periyar Lake

- **Cochin Port:** Natural Port under Ministry of Shipping.
- **Kollam and Vizhinjam Ports:** Under Kerala Government.

ARABIAN SEA

LAKSHADWEEP

NATIONAL WATERWAY

Kottappuram

Kochi

Cherthala

Alappuzha

Sasthamkotta Lake

Kollam

Anjarakandy Kalim

Chaliyar

Kunthi

Bharathapuzha

Chalakydy

periyar

Kaliyar

Thodupuzha

Ieenachil

Manimala

Pampa

Achanthovil

Kallada

Itakkara

Neyyar

- Declared in 1993 and stretches from Kollam to Kottapuram of West Coast Canal and Champakara and Udyogamandal Canals.
- Recently, Centre has turned down a proposal from the Kerala government to extend the NW3 from Kollam to Kovalam and Kozhikode to Bekal.

Periyar Lake

SRI LANKA

WILDLIFE SANCTUARIES



- **Chinnar WLS:** Earned the name for only rehabilitation centre for Indian star tortoise.
- **Neyyar WLS:**
 - ✓ Drainage basin for the Neyyar River and its tributaries.
 - ✓ Agasthyamalai is a very prominent landmark.
 - ✓ Kochi high court ordered that women cannot be excluded from trekking to Agasthyarkoodam. It is the second highest peak in Kerala situated in this WLS.

LAKSHADWEEP

WLS - WILDLIFE SANCTUARY

SRI LANKA



ARABIAN SEA

BAY OF BENGAL

KARNATAKA

TAMIL NADU

VEMBANAD LAKE

SASTHAMKOTTA LAKE

ASHTAMUDI LAKE

RW - RAMSAR WETLANDS

- **Vembanad - Kol Wetland (VKW):** Largest estuarine system of the western coastal wetland systems.
- **Sasthamkotta Lake (SL):** Largest freshwater lake in Kerala.

STATE BY STATE ... KERALA

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BIOSPHERE RESERVES



NILGIRI BR

Nilgiri BR:

- ✓ First biosphere reserve in India
- ✓ Established in 1986.
- ✓ Part of World Network of Biosphere Reserves.

Agasthyamalai BR:

- ✓ Part of World Network of Biosphere Reserves.

AGASTHYAMALAI BR

BR - BIOSPHERE RESERVES



KARNATAKA

ARABIAN SEA

BAY OF BENGAL

TAMIL NADU

LAKSHADWEEP

Parambikulam TR

Periyar TR

TR - TIGER RESERVE

- Both the tiger reserves attained good ranking in management effectiveness evaluation in recent Tiger Census.
- Suffer from porous borders, illegal entry points and pressure from tourism like Sabarimala etc.

STATE BY STATE ... KERALA

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ELEPHANT RESERVES (ER)

BRAHMAGIRI-NILGIRI-EASTERN
GHAT LANDSCAPE

MYSORE ER
KARNATAKA

WAYANAD ER
KERALA

NILGIRI ER
TAMIL NADU

NILAMBUR ER
KERALA

COIMBATORE ER
TAMIL NADU

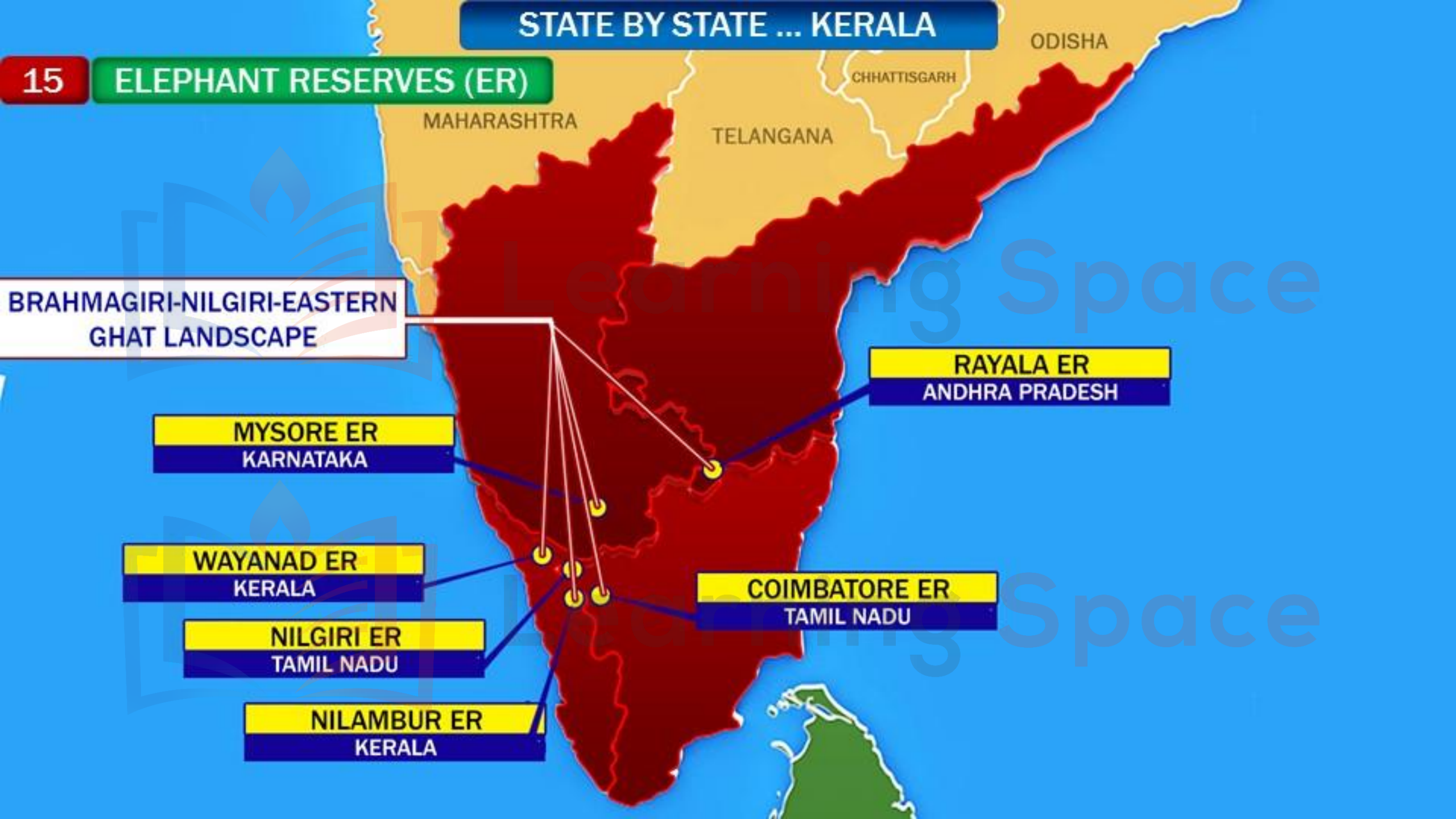
RAYALA ER
ANDHRA PRADESH

MAHARASHTRA

TELANGANA

CHHATTISGARH

ODISHA



ANAMALAI-NELLIAMPATHY-
HIGH RANGE LANDSCAPE

ANAMUDI ER
KERALA

ANAMALAI ER
TAMIL NADU

ANDHRA PRADESH

KARNATAKA

Learning Space

Learning Space

ANDHRA PRADESH

KARNATAKA

PERIYAR-AGASTHYAMALAI
LANDSCAPE

SRIVILLIPUTHUR ER
TAMIL NADU

PERIYAR ER
KERALA

Learning Space

Learning Space

STATE BY STATE ... KERALA

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UNESCO - WHS



Western Ghats:

- UNESCO World Heritage Site and is one of the eight "hottest hot-spots" of biological diversity in the world.
- A total of 39 areas in the Western Ghats, including national parks, wildlife sanctuaries and reserve forests, were designated as world heritage sites in 2012.

KATHAKALI DANCE



- Blend of dance, music and acting and dramatizes stories, which are mostly adapted from the Indian epics.
- Four aspects of abhinaya - angika, aharya, vachika, satvika and the nritya, nritya and natya aspects are combined perfectly.
- Derives its textual sanction from Balarama Bharatam and Hastalakshana Deepika.

MOHINIYATTAM DANCE



- Literally interpreted as the dance of 'Mohini', the celestial enchantress of the Hindu mythology.
- Classical solo dance form performed by women.
- Characterized by graceful, swaying body movements with no abrupt jerks or sudden leaps.
- Belongs to the lasya style which is feminine, tender and graceful.

MUDIYETTU



- Inscribed on the Representative List of the Intangible Cultural Heritage of Humanity in 2010.
- Usually performed only in the Kali temples of Kerala, as an oblation to the Goddess.
- Depicts the triumph of goddess Bhadrakali over the asura Darika.

KALARIPAYATTU



- An Indian martial art and fighting system that originated in modern-day Kerala.
- Practitioners of Kalaripayattu possess intricate knowledge of pressure points on the human body and healing techniques that incorporate the knowledge of Ayurveda and Yoga.

THEYYAM



- Performers of Theyyam belong to the community which are in below level in ancient caste structure formed by Namboothiri brahmins in Kerala.
- Performed mainly by males, except the Devakoothu theyyam.
- Devakoothu Theyyam ritual performed only in the Thekkumbad Kulom temple.

PAVAKOOTHU



- Traditional glove puppet play is came into existence during the 18th century due to the influence of Kathakali.
- Head and the arms are carved of wood and joined together with thick cloth, cut and stitched into a small bag.
- Theme for Glove puppet plays is based on the episodes from either the Ramayana or the Mahabharata.

KUTIYATTAM



- Inscribed on the Representative List of the Intangible Cultural Heritage of Humanity in 2008.
- Kutiyattam is traditionally performed by Chakyars in theatres called Kuttampalams.
- Women of Nambiar caste play the female roles.
- Emphasis on hand gestures and eye movements makes this dance and theatre form unique.

ONAM FESTIVAL



- State festival of Kerala
- Observed by all its citizens, irrespective of caste, creed or community.
- Harvest festival celebrated during the Malayalam month of Chingom (August-September).
- Vallamkali or Snake Boat Races is the traditional boat race that marks the end of Onam festival.

POORAM FESTIVAL



- Annual festival celebrated in temples dedicated to goddesses Durga or Kali after the summer harvest.
- Thrissur Pooram is celebrated mainly to worship Lord Shiva and has a lot of colour and music.

PVTGs in Kerala: Cholanaikayan (a section of Kattunaickans), Kadar, Kattunayakan, Kurumbas, Koraga.

CHOLANAICKAN TRIBE



- Found mainly in Silent Valley NP.
- One of the last remaining hunter-gatherer tribes of South India
- Primarily speak the Dravidian Cholanaikkan language.
- Live in rock shelters called 'Kallulai' or in open campsites made of leaves.
- Found in groups each called a 'Chemmam'

KADAR TRIBE



- Live in the forests and have not been observed to practice any form of agriculture.
- As many as 23 families of Kallar Kadar tribal community were evicted from their settlement inside Anamalai Tiger Reserve (ATR).

KURUMBA TRIBE



- Hamlets are found in both sides of the Bhavani River in Attapady area of Palakkad district.
- Most of them are still food gatherers and also do shift cultivation.
- A hamlet is called Ooru and the head of the ooru is called Oorumooppan, who administers the socio-political unit.

GREAT HORNBILL



- State bird of Kerala.
- Hunting poses a substantial threat to the species.
- River Chalakudy is the lifeline for the bird population.
- Placed under Schedule 1 of the Wildlife Protection Act, 1972.
- IUCN Status: Vulnerable

NILGIRI WOODPIGEON



- Endemic to the hill-ranges of the Western Ghats.
- Species is suspected to be declining at a moderate rate, based on on-going rates of habitat loss and potential hunting pressure.
- IUCN Status: Vulnerable

MALABAR CIVET



- Endemic to the Western Ghats
- Listed in Schedule I, Part I of the Wildlife (Protection) Act, 1972 and on CITES Appendix III.
- Deforestation and commercial plantations are major threats.
- IUCN Status: Critically Endangered.

NILGIRI LANGUR



- Also known as the hooded leaf monkey.
- Hunted for its skin, which is used for making drums, as well as for other parts of the body, which are used for meat as well as in traditional "medicine"
- 'Mason-Pfizer Monkey Virus (M-PMV)' is highly vulnerable among Nilgiri Langurs.
- IUCN Status: Endangered.

HUMP-BACKED MAHSEER



- Species of freshwater ray-finned fish.
- Referred to as tiger of the water.
- Found only in the Cauvery river basin including Pambar, Kabini and Bhavani rivers.
- Shoal is an international organization working to conserve freshwater species.
- It initiated 'Project Mahseer' in collaboration with other stakeholders.
- IUCN Status: Critically Endangered.

MEWA SINGH'S NIGHT FROG



- Discovered in Kozhikode's Malabar Wildlife Sanctuary.
- Endemic only to Western Ghats mountain range.
- Named in honour of wildlife scientist Mewa Singh for his contributions to behavioural ecology and primate studies.

DECOLOURISATION OF RIVER PERIYAR



- Pollution control Board (PCB) has attributed the change in colour to the poor quality of water as a result of eutrophication.
- PCB has stated that huge quantities of organic load in the form of sewage from nearby townships are regularly reaching the river system.
- This is contributing to the eutrophication resulting in the change of colour.

KUDUMBASHREE MISSION



- Envisioned as a part of the People's Plan Campaign and local self-governance, with women at the centre of it.
- Played significant role in relief operations in recent Kerala floods.
- It has a three-tier structure:
 - ✓ Neighbourhood groups (NGs);
 - ✓ Area development societies (ADS); and
 - ✓ All ADSs are federated through the community development societies (CDS).

MARAYUR JAGGERY



- Marayoor in the Idukki district of Kerala is known for its extensive sugarcane cultivation.
- Peculiar geographical location of Marayoor amid the forests of the Western Ghats gives the sugar cane a distinct geographical identity.
- High sweetness with less saltiness, high content of iron and less sodium.
- Produce is free of impurities and the sugar cane fields are free of chemical pesticides and fertilizers.

TIRUR BETEL LEAF



- Valued both for its mild stimulant action and medicinal properties.
- Commonly used for making pan masala for chewing, it has many medicinal, industrial and cultural usages.
- Considered as a remedy for bad breath and digestive disorders.

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

0866 244 44 72

98499 42299

94415 27588